

HDI Predictor Documentation

Project Initialization and Planning Phase — Project Proposal (Proposed Solution)

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Team ID	SB-GENAI-HDI-2025
Project Title	HDI Predictor - Human Development Index Classification System
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The proposal aims to make human development assessment instant and accessible by replacing manual HDI computation with a trained machine learning classifier. It tackles the inefficiency of manually cross-referencing indicator tables, promising faster, consistent, and explainable development-tier classification. Key features include a RandomForest-based tier classifier, an HDI score estimate, and a lightweight web interface.

Project Overview

Field	Details
Objective	Build and deploy a machine learning system that classifies a country (or hypothetical scenario) into a Human Development tier — Very High, High, Medium, or Low — from its life expectancy, education, and income indicators.
Scope	Covers data preparation, model training and evaluation, a Flask-based prediction interface, and Docker packaging for reproducible deployment.

Problem Statement

Field	Details
Description	Manually deriving a country’s development tier from raw HDI indicators is slow, inconsistent, and inaccessible to non-statisticians.
Impact	Solving this reduces the time researchers and policymakers spend on routine classification, freeing them to focus on interpreting results and shaping interventions.

Proposed Solution

Field	Details
Approach	Train a supervised classification model (RandomForestClassifier) on labeled country-level HDI data, then serve it through a Flask API/web form.
Key Features	(1) Instant tier classification from four core indicators. (2) Underlying HDI score estimate. (3)

	Feature-importance summary for explainability. (4) Dockerized deployment for one-command reproducibility.
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Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	CPU used for model training (RandomForest does not require GPU)	Standard CPU, 4 cores
Memory	RAM for training and serving	8 GB
Storage	Disk space for dataset, model artifact, and logs	5 GB
Frameworks	Python web framework	Flask
Libraries	ML and data libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE / notebook	Jupyter Notebook, VS Code
Data	Source, size, format	Country-level HDI indicator dataset, CSV, ~190 country rows across historical years
Deployment	Containerization	Docker